Computational Psychodynamics: A New Framework for Objective Cognitive Assessment

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The author is the founder and president of Beverly Index LLC, a company developing commercial applications based on the theoretical framework presented in this paper. This potential conflict of interest is disclosed in accordance with APA ethical standards. The theoretical framework presented herein is offered for scientific scrutiny and empirical testing by the research community.

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Abstract

This paper introduces Computational Psychodynamics, a proposed scientific discipline concerned with the extraction of cognitive and psychological information from the structural properties of linguistic output. The framework rests on a central premise: that internal cognitive states—shaped by neural oscillatory activity across distributed brain networks—systematically influence language production in ways that leave detectable statistical signatures in spoken and written text. Rather than relying on self-report measures or subjective clinical judgment, Computational Psychodynamics proposes that careful computational analysis of lexical, syntactic, semantic, temporal, and prosodic features of language can yield objective indices of cognitive function. The present paper articulates the theoretical foundations of this discipline, situates it within the broader landscape of psycholinguistics, cognitive neuroscience, and computational psychiatry, identifies the key scientific problems it aims to address, and proposes a research agenda for empirical validation. Two core theoretical contributions are introduced: Neural Oscillation Signature Theory (NOST), which formalizes the proposed relationship between oscillatory states and linguistic output, and the Neural Oscillatory Spectroscopy System (NOSS), a 12-sector classification taxonomy mapping cognitive domains to their predicted oscillatory–linguistic signatures. The framework is offered as a theoretical contribution warranting empirical investigation rather than an established finding.

Keywords: computational psychodynamics, cognitive assessment, neural oscillations, language production, psycholinguistics, computational psychiatry, NOST, NOSS

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Computational Psychodynamics: A New Framework for Objective Cognitive Assessment

The assessment of cognitive and psychological states remains one of the most consequential yet methodologically constrained problems in the behavioral sciences. Despite remarkable advances in neuroimaging, electrophysiology, and computational modeling, the dominant instruments for evaluating mental health and cognitive function continue to rely heavily on subjective self-report, clinician judgment, and structured interviews—methods whose limitations have been extensively documented (Fried, 2017; Kotov et al., 2017). These limitations are not trivial: diagnostic concordance between clinicians assessing the same patient frequently falls below acceptable thresholds (Regier et al., 2013), and the average delay from symptom onset to accurate psychiatric diagnosis exceeds several years in many conditions (Wang et al., 2005).

This paper proposes that a significant and underexploited source of diagnostic information exists in the structural properties of natural language. The central claim is not merely that language reflects thought—a truism dating to antiquity—but rather that the specific statistical features of linguistic output carry extractable information about the neural oscillatory processes that generated them. If this claim is correct, then careful computational analysis of language could provide a non-invasive, scalable, and objective method for inferring the cognitive states of speakers and writers.

We term this proposed scientific discipline Computational Psychodynamics: the systematic study of how cognitive states encode in, and can be inferred from, the structural properties of linguistic output. The name reflects both its computational methodology and its

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concern with the dynamic processes—neural, cognitive, and linguistic—that connect internal states to external expression.

It is important to state at the outset what this paper is and what it is not. This is a theoretical contribution. It proposes a framework, derives testable predictions, and identifies the research program necessary to validate or refute those predictions. It does not present empirical results, nor does it claim that the proposed framework has been validated. We offer it to the scientific community for critical scrutiny.

The Problem: Subjectivity in Cognitive Assessment

The current landscape of cognitive and psychological assessment is characterized by a fundamental tension: the phenomena being measured are among the most consequential in human experience—mood, anxiety, psychosis, cognitive decline, trauma response—yet the instruments used to measure them remain largely dependent on subjective processes. This section examines the specific limitations that motivate the search for new assessment approaches.

Limitations of Current Assessment Methods

Contemporary psychological assessment relies on four primary methodologies, each with well-documented constraints. Self-report instruments, such as the PHQ-9 for depression and the GAD-7 for anxiety, are subject to social desirability bias, limited insight, and cultural variation in symptom reporting (Paulhus, 1991; Choi & Nisbett, 1998). Structured clinical interviews, while more rigorous, introduce clinician variability: the DSM-5 field trials found that kappa values for several major diagnostic categories fell below 0.5, indicating that clinicians frequently disagree on the same patient (Regier et al., 2013).

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Neuropsychological testing provides more standardized measurement but is typically constrained to specific cognitive domains, requires trained administrators, and captures performance at a single point in time rather than ongoing cognitive function. Neuroimaging offers biological measurement but remains prohibitively expensive for routine clinical use, requires specialized equipment and expertise, and has not yet produced diagnostic biomarkers with sufficient sensitivity and specificity for individual-level clinical decision-making (Woo et al., 2017).

These limitations create a diagnostic bottleneck that has real consequences. The average time from onset of psychotic symptoms to treatment initiation exceeds one year in most health systems (Marshall et al., 2005). Early-stage neurodegenerative conditions, including Alzheimer's disease and Parkinson's disease, progress for years before current assessment methods detect them (Jack et al., 2018; Postuma et al., 2012). Traumatic brain injury often goes undetected when structural imaging appears normal despite measurable cognitive impairment (Bigler, 2013).

Language as an Untapped Diagnostic Resource

Language is the most information-rich behavioral output humans produce. A typical adult generates thousands of words per day, each reflecting choices—largely unconscious—among vast combinatorial possibilities. Research across psycholinguistics, computational psychiatry, and natural language processing has increasingly demonstrated that these choices are not random but are systematically influenced by the speaker's cognitive and affective state.

Studies have shown that depression alters lexical selection toward increased first-person singular pronoun use and negative emotion words (Rude et al., 2004; Pennebaker et al., 2003). Psychotic disorders produce measurable reductions in semantic coherence and increases in tangential discourse (Elvevåg et al., 2007; Bedi et al., 2015). Neurodegenerative conditions

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produce progressive simplification of syntactic complexity and reductions in vocabulary diversity (Fraser et al., 2016; Garrard et al., 2005). Post-traumatic stress disorder generates characteristic patterns of fragmented temporal reference and avoidant language surrounding trauma-related content (Jaeger et al., 2014).

These findings, while compelling, remain largely disconnected. No comprehensive theoretical framework currently exists for understanding why different cognitive states produce different linguistic signatures, what the mechanistic basis of these associations is, or how they might be systematically exploited for diagnostic purposes. Computational Psychodynamics proposes to provide such a framework.

Theoretical Foundations

Computational Psychodynamics rests on three foundational pillars, each grounded in established empirical findings but integrated in a novel theoretical synthesis.

Pillar 1: Neural Oscillatory Governance of Cognition

Neural oscillations—rhythmic fluctuations in electrical activity across brain networks—are among the most robust and replicated findings in cognitive neuroscience. They are not epiphenomenal but play functional roles in organizing cognitive processes including attention (Fries, 2005; Jensen & Mazaheri, 2010), memory encoding and retrieval (Hanslmayr et al., 2016; Nyhus & Curran, 2010), executive control (Cavanagh & Frank, 2014), and emotional regulation (Knyazev, 2007). Different frequency bands—delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–100 Hz)—are associated with distinct cognitive operations, and the coupling between bands reflects cross-regional coordination (Canolty & Knight, 2010).

Critically for the present framework, psychiatric and neurological conditions are associated with characteristic disruptions to oscillatory activity. Schizophrenia involves reduced

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gamma-band synchrony (Uhlhaas & Singer, 2010). Depression is associated with frontal alpha asymmetry (Thibodeau et al., 2006). Parkinson's disease involves pathological beta-band oscillations in basal ganglia–cortical circuits (Little & Brown, 2014). Traumatic brain injury disrupts cross-frequency coupling patterns (Palacios et al., 2019). These disruptions are not merely correlates of pathology; they reflect altered computational properties of the neural circuits that generate cognition.

Pillar 2: Cognitive Determination of Linguistic Output

The second pillar draws on the psycholinguistic principle that language production is not an independent process but is constrained by—and therefore reflects—the cognitive operations that feed it. Levelt's (1989) foundational model of speech production demonstrates that the progression from conceptualization through formulation to articulation is shaped at every stage by cognitive resources including working memory, attentional allocation, and executive control. Bock and Levelt (1994) showed that syntactic choices during sentence production reflect the accessibility of competing representations, which is itself a function of underlying cognitive processes.

More recent work has extended this principle to computational models. Jaeger (2010) demonstrated that speakers optimize information density in ways that reflect cognitive processing constraints. MacDonald (2013) showed that production choices are shaped by the producer's experience and cognitive biases. The implication is straightforward: if cognitive states shape language production, and if different cognitive states produce different linguistic patterns, then language output carries extractable information about the cognitive processes that generated it.

Pillar 3: Computational Extraction of Latent Structure

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The third pillar is methodological: advances in natural language processing (NLP) and machine learning have created the technical capacity to extract subtle statistical regularities from linguistic data that are invisible to human observation. Modern NLP techniques can quantify semantic coherence through embedding similarity (Mikolov et al., 2013), measure syntactic complexity through parse tree analysis (Lu, 2010), assess temporal dynamics through discourse analysis (Graesser et al., 2004), and detect prosodic features through acoustic signal processing (Eyben et al., 2013).

These three pillars—oscillatory governance of cognition, cognitive determination of language, and computational extraction of latent structure—constitute the theoretical spine of Computational Psychodynamics. Their integration yields the framework's central theoretical claim.

The Central Theoretical Claim

Computational Psychodynamics proposes the following: Neural oscillatory patterns systematically shape cognitive processing, which in turn systematically shapes linguistic output. Therefore, the statistical properties of linguistic output carry extractable information about the neural oscillatory configuration that generated them. It is worth noting that this epistemological structure—inferring the properties of hidden generative states from the statistical regularities of their observable output—is not unique to the cognitive-linguistic domain. It is the same inferential strategy employed by spectroscopy, seismology, and gravitational wave astronomy, among other fields, and recent findings in astrophysics and particle physics continue to demonstrate its productivity across scales (see Beverly, 2026b, for extended discussion). This claim can be expressed more precisely through Neural Oscillation Signature Theory.

Neural Oscillation Signature Theory (NOST)

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NOST formalizes the central claim of Computational Psychodynamics into a testable theoretical framework. Its core proposition is that each distinguishable cognitive state—whether healthy or pathological—produces a characteristic pattern of neural oscillatory activity that leaves a detectable "signature" in the linguistic output generated under that state. The theory proposes a causal chain: oscillatory state → cognitive processing configuration → linguistic feature profile.

NOST makes several specific predictions. First, that linguistic features can be organized into classes corresponding to the oscillatory bands that primarily influence them: delta-range processes are predicted to influence narrative coherence and thematic integration; theta-range processes are predicted to influence memory-related lexical features and temporal reference patterns; alpha-range processes are predicted to influence attentional markers and information selection; beta-range processes are predicted to influence syntactic planning and motor-speech coordination; and gamma-range processes are predicted to influence semantic binding and perceptual description fidelity.

Second, NOST predicts that clinical conditions characterized by known oscillatory disruptions should produce linguistic signatures that correspond to those disruptions. For example, the pathological beta-band activity in Parkinson's disease should manifest as measurable changes in both motor-speech characteristics (articulation rate, voice quality) and the cognitive-linguistic features associated with basal ganglia–cortical circuit function (verb complexity, action language, sequential planning in discourse).

Third, NOST predicts that linguistic signatures should change in response to effective treatment, tracking with changes in the underlying oscillatory pathology. This prediction, if confirmed, would provide a basis for objective treatment monitoring.

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A companion paper (Beverly, 2026b) provides the complete theoretical development and mathematical formalization of NOST.

The Neural Oscillatory Spectroscopy System (NOSS)

NOSS operationalizes NOST by organizing the space of cognitive states into a 12-sector classification system, each sector corresponding to a clinically meaningful domain of cognitive function with its own predicted oscillatory–linguistic signature profile. The 12 sectors are: (S1) Mood, encompassing affective regulation and emotional baseline stability; (S2) Anxiety, encompassing threat-response, hypervigilance, and autonomic arousal; (S3) Psychotic, encompassing reality-processing integrity and thought organization; (S4) Neurocognitive, encompassing executive function, memory, and attention; (S5) Developmental, encompassing ADHD, autism spectrum, and neurodevelopmental processing differences; (S6) Optimal, representing healthy cognitive baseline and resilience; (S7) Traumatic Brain Injury, encompassing concussion and post-concussive syndrome; (S8) Neurodegenerative, encompassing progressive conditions including Parkinson's disease; (S9) Captivity Trauma, encompassing coercion, identity suppression, and complex PTSD resulting from prolonged captivity; (S10) Substance, encompassing addiction and reward-system distortion; (S11) Dissociative, encompassing fragmentation, depersonalization, and derealization; and (S12) Personality, encompassing enduring interpersonal patterns and identity stability.

Each sector is defined by its predicted oscillatory disruption profile, the cognitive processes those disruptions are expected to affect, and the linguistic features those cognitive changes are expected to produce. NOSS thereby translates the abstract theoretical claims of NOST into a concrete classification system amenable to empirical investigation. A dedicated

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paper (Beverly, 2026c) provides the complete development of NOSS with full sector specifications.

Relationship to Existing Disciplines

Computational Psychodynamics does not claim to be constructed in isolation. It draws on and integrates findings from several established fields, while proposing a distinctive theoretical synthesis.

Computational Psychiatry

Computational psychiatry applies mathematical and computational modeling to understand psychiatric phenomena (Huys et al., 2016). Computational Psychodynamics shares this commitment to quantitative rigor but differs in its focus: where computational psychiatry has primarily modeled internal cognitive processes (e.g., reinforcement learning disruptions in addiction, aberrant prediction error signaling in psychosis), Computational Psychodynamics focuses on the external behavioral signal—linguistic output—as a window into those processes. The two approaches are complementary rather than competing.

Natural Language Processing in Mental Health

An active research literature applies NLP methods to mental health detection, particularly using social media data (De Choudhury et al., 2013; Coppersmith et al., 2015). Computational Psychodynamics differs from this work in two respects. First, it proposes a theoretical framework—grounded in neural oscillatory mechanisms—that explains why linguistic features should vary with mental health status, rather than relying on atheoretical feature engineering. Second, it proposes a comprehensive classification system (NOSS) that extends beyond the conditions most commonly studied in NLP-based mental health research.

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Psycholinguistics and Cognitive Science

The psycholinguistic literature provides the empirical grounding for Pillar 2 of the framework. Computational Psychodynamics extends this work by proposing specific mappings between neural oscillatory bands and classes of linguistic features—mappings that, if validated, would provide a deeper mechanistic understanding of how cognitive states influence language production.

Proposed Application Domains

If the theoretical claims of Computational Psychodynamics are empirically validated, the framework implies several potential application domains. We describe these briefly while emphasizing that they remain speculative pending validation.

Clinical Cognitive Assessment

The most direct application would be the development of language-based cognitive assessment instruments. Under this pathway, standardized narrative samples—dream reports, autobiographical narratives, structured interview responses—would be computationally analyzed to produce objective indices of cognitive function across the 12 NOSS sectors. Such instruments could potentially complement existing assessment methods by providing quantitative data derived from naturalistic behavior rather than structured test performance.

Narrative Authenticity Analysis

NOST's predictions imply that authentic recall and deliberate fabrication should produce distinguishable linguistic signatures, because the cognitive processes underlying genuine memory retrieval differ from those underlying narrative construction (Vrij et al., 2019). The framework thus suggests potential forensic applications in assessing narrative authenticity,

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though such applications would require rigorous validation against established standards of evidence.

Longitudinal Cognitive Monitoring

Because linguistic output can be sampled repeatedly and non-invasively, the framework suggests applications in longitudinal monitoring—tracking cognitive changes over time in response to treatment, aging, or neurological progression. The theoretical integration of Psychological Lense-Thirring Precession—a model describing how significant psychological experiences produce observable oscillatory perturbations in linguistic output that dampen as cognitive integration proceeds—provides a specific quantitative framework for predicting how linguistic signatures should evolve over time during psychological processing.

Limitations and Requirements for Empirical Validation

The framework presented here is theoretical. While it integrates established findings from cognitive neuroscience, psycholinguistics, and computational linguistics, the specific claims of NOST and the classification system of NOSS remain to be empirically tested. Several critical requirements must be met before the framework can be considered validated.

First, the predicted correspondences between oscillatory bands and linguistic feature classes require concurrent measurement—simultaneous EEG recording and language sample collection—to establish whether the proposed mappings hold. Second, the NOSS sector signatures must be validated against criterion measures: clinical diagnoses established through gold-standard assessment methods. Third, the framework's predictions regarding treatment response must be tested in longitudinal designs tracking both linguistic signatures and clinical outcomes over time.

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Fourth, significant questions of generalizability remain. The relationship between cognitive states and linguistic output may be moderated by language, culture, education, age, and other demographic variables. The framework's clinical utility depends on demonstrating that its predictions hold across diverse populations. Fifth, the distinction between state effects (temporary cognitive configurations) and trait effects (enduring processing styles) requires careful investigation, as the framework in its current form does not fully specify how these interact in linguistic output.

Finally, it should be acknowledged that the author's commercial interest in applications of this framework creates a potential bias that independent investigators should consider when evaluating the theoretical claims. The author has attempted to present the framework with appropriate epistemic humility, but independent replication and critique are essential to the scientific process.

Conclusion

Computational Psychodynamics proposes that the structural properties of linguistic output carry extractable information about the cognitive states of speakers and writers, information rooted in the neural oscillatory processes that govern cognition and shape language production. The framework integrates established findings from cognitive neuroscience, psycholinguistics, and computational linguistics into a novel theoretical synthesis that generates specific, testable predictions.

Two core theoretical contributions have been introduced: NOST, which formalizes the proposed relationship between oscillatory states and linguistic signatures, and NOSS, which organizes the space of cognitive states into a 12-sector classification system with predicted

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oscillatory–linguistic profiles. Together, these provide a research program amenable to systematic empirical investigation.

Whether the specific claims of this framework prove correct is an empirical question that cannot be settled by theoretical argument alone. What can be stated with reasonable confidence is that the existing evidence—from cognitive neuroscience, psycholinguistics, computational psychiatry, and NLP-based mental health research—is sufficient to warrant taking the framework's predictions seriously and investigating them rigorously. The scientific community's scrutiny is both welcomed and necessary.

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